

STUDY GUIDE FOR MATH 195 MIDTERM 1

TYPES OF QUESTIONS

The exam has 15 questions. Expect to see a few conceptual questions, with most questions being problem-solving questions.

- **Conceptual questions** ask you to explain something. These are writing questions.
- **Problem-solving questions** ask you to determine or compute something. They might ask you to sketch or interpret a graph. Or they might ask you to algebraically determine some datum about a function.

TOPICS TO KNOW

So far this semester we've looked a few core concepts which can be applied to many different kinds of functions.

- How to graph functions and how to use geometric transformations to determine the behavior of graphs.
- How to get big picture information such as domain from a graph.
- How to algebraically determine information about functions, such as intercepts, intersections, or inverses.
- The algebraic and geometric meaning of inverses. How to use inverses to solve equations.

You should understand how to apply these ideas to the following classes of functions.

- Linear;
- Quadratic;
- Power;
- Reciprocal of power;
- Radical;
- Exponential;
- Logarithm.

Additionally, you should know how to apply some special algebraic properties.

- Solving quadratic equations.
- Rules for exponents.
- Rules for logarithms.

SAMPLE QUESTIONS

- (1) Find the intersection point of the two lines

$$y = 3x - 2 \quad \text{and} \quad y = -x + 1$$

- (2) Find the x - and y -intercepts of $a(x) = 2x^2 - 8x + 5$.
- (3) Find the vertex of $b(x) = -x^2 - 6x + 3$.
- (4) Sketch a graph of $c(x) = -2(x - 4)^2 + 3$, identifying its vertex. State its domain and range, and where it is increasing or decreasing. Is it concave up or concave down?
- (5) Sketch a graph of $d(x) = 4x^2 + 2x + 1$, identifying its vertex. State its domain and range, and where it is increasing or decreasing. Is it concave up or concave down?
- (6) Find all points where the parabolas $y = x^2 - 4$ and $y = (x - 4)^2$ intersect.
- (7) Sketch a graph of $f(x) = -x^4 - 3$, identifying its vertex. State its domain and range, and where it is increasing or decreasing. Is it concave up or concave down?
- (8) Sketch a graph of $g(x) = 2(x + 1)^3 - 2$, identifying its inflection point. State its domain and range, and where it is increasing or decreasing. Is it increasing or decreasing?
- (9) Find all intercepts of $g(x) = 2(x + 1)^3 - 2$.

- (10) Give a formula for the inverse of $g(x) = 2(x+1)^3 - 2$.
- (11) Sketch a graph of $h(x) = \frac{3}{x+2} - 1$, identifying its asymptotes. State its domain and range, and where it is concave up or concave down. Where is it increasing? Decreasing?
- (12) Determine a formula for the inverse of $j(t) = \frac{3}{(t-1)^5} + 4$.
- (13) Sketch a graph of $k(x) = -\sqrt{2-x}$. What are its domain and range?
- (14) Sketch a graph of $\ell(x) = -\sqrt[5]{3+x} - 4$. What are its domain and range? Where is it concave up or concave down? Is it increasing or decreasing?
- (15) Find all points where the functions $m(x) = x$ and $n(x) = \sqrt{1-x}$ intersect.
- (16) Find all intercepts of $p(x) = 2\sqrt[3]{2x-4} + 4$.
- (17) Give a formula for the inverse of $q(x) = \sqrt[3]{4-3x} - 3$.
- (18) Sketch a graph of $r(x) = 4e^{2-x} + 1$, identifying the asymptote. What are the domain and range? Is it increasing or decreasing? Concave up or concave down?
- (19) Sketch a graph of $s(x) = -(\frac{1}{\pi})^{2x} + 3$, identifying the asymptote. What are the domain and range? Is it increasing or decreasing? Concave up or concave down?
- (20) Find all points of intersection of $t(x) = 3^x$ and $u(x) = 2^{4-x}$.
- (21) Find all intercepts of $v(x) = 4 - 2^{3-x}$.
- (22) Rewrite the following formula to have a single logarithm.

$$\log_5(x+1) - 2\log_5(x) + \log_5(25) + 3\log_5(x-1)$$

- (23) Rewrite the following formula to be a sum/difference of multiple logarithms, where the input to each logarithm is as simple as possible.

$$\log_{10} \left(\frac{2t(t-1)^2}{(t+1)(t+2)^3} \right)$$

- (24) Sketch a graph of $w(x) = -\ln(2-x)$, identifying its asymptote. What are the domain and range? Is it increasing or decreasing? Concave up or concave down?
- (25) Find all intercepts of $y(x) = 4 - 2\log_2(6-2x)$.
- (26) Find all x -intercepts of $z(x) = \log_3(x^2 + 5x + 5)$.